

ITELEC 3208: Data Scalability and Analytics

2nd Semester A.Y. 2024-2025

PROJECT – Python Data Analytics

Members: Jeric J. Dela Cruz Franz Harry Daniel A. Eda	Score:
Section: BSIT 3-2	Date:

I. Directions. *Follow the instructions properly as listed below. Any mistakes will result in grade deduction so be careful.*

Requirements

1. Determine the topic/target of your data analytics study.
2. The team must gather or extract existing data.
3. The team will proceed with processing data (cleaning and formatting).
4. Provide data analytics and visualization of your process data using Python.
5. Documentation.

Project Submission

1. The documentation and source code must be submitted on or before the set deadline.

Rubrics - Refer to this table for a reference regarding the project's grading system:

#1 – Documentation		Scores		
Category	Specifications	Excellent	Good	Poor
Difficulty of Scope	The broadness, relevance and difficulty of the project's scope.	25	15	10
Documentation	The document is easy and simple to understand; the system is explained properly through proper grammar; terms are used and well-organized in a professional manner.	25	15	10
Coding Standard	Coding is properly done complete with indentations, professional and proper variable, class, etc. naming and conventions are used, and the overall flow is easy to follow and understand.	30	25	15
Outputs	Correct usage of graphs and easy to understand.	20	15	5
Maximum Score		100	70	40
Total		100		

Prepared by:

Santi S. Ognita
Instructor I

Approved by:

Khavée Agustus W. Botangen
Chair, Department of Information Technology

Students' Social Media Addiction Dataset Documentation

Overview

This dataset contains survey responses from students aged 16 to 25 across multiple countries. It captures their social media usage patterns alongside key personal, academic, and well-being indicators. The dataset is structured for easy import into statistical software or data science environments, including both quantitative and categorical fields. It supports exploratory analysis, predictive modeling, and reporting on the relationships between social media addiction and various life outcomes.

Dataset Scope & Coverage

- Population:** Students aged 16–25 enrolled in high school, undergraduate, or graduate programs.
- Geography:** Multi-country coverage, including Bangladesh, India, USA, UK, Canada, Australia, Germany, Brazil, Japan, and South Korea.
- Timeframe:** Data collected via a one-time online survey administered in Q1 2025.
- Volume:** Sample sizes configurable by research requirements, from 100 to over 10,000 respondents.

Data Structure & Column Descriptions

Column Name	Type	Description
Student_ID	Integer	Unique respondent identifier, used for de-duplication and record tracking without revealing identity.
Age	Integer	Student's age in completed years at survey time.
Gender	Categorical	Self-reported gender ("Male" or "Female").
Academic_Level	Categorical	Current highest education level ("High School", "Undergraduate", "Graduate").
Country	Categorical	Country of residence where survey was completed.
Avg_Daily_Usage_Hours	Float	Average number of hours spent daily on social media platforms, derived from weekday and weekend estimates.
Most_Used_Platform	Categorical	Primary social media platform used (e.g., Instagram, Facebook, TikTok).
Affects_Academic_Performance	Boolean	Self-reported perception of whether social media negatively impacts academic performance ("Yes"/"No").
Sleep_Hours_Per_Night	Float	Average nightly sleep duration in hours.
Mental_Health_Score	Integer	Self-rated mental well-being on a scale of 1 (poor) to 10 (excellent).
Relationship_Status	Categorical	Current romantic relationship status ("Single", "In Relationship", "Complicated").
Conflicts_Over_Social_Media	Integer	Number of reported conflicts with family, friends, or partners due to social media use.
Addicted_Score	Integer	Composite score (1 low to 10 high) based on a standardized social media addiction scale (e.g., Bergen Social Media Addiction Scale).

Documentation:

For this project, we chose to use Kaggle and Jupyter Notebook because they provide a powerful and flexible environment for data analysis and visualization. These tools allow us to easily explore the dataset, create clear and interactive charts, and present insights in a way that is easy to understand. Using Jupyter Notebook on Kaggle enables us to combine code, visualizations, and explanations all in one place, making it ideal for showcasing the key findings of our study on students' social media addiction.

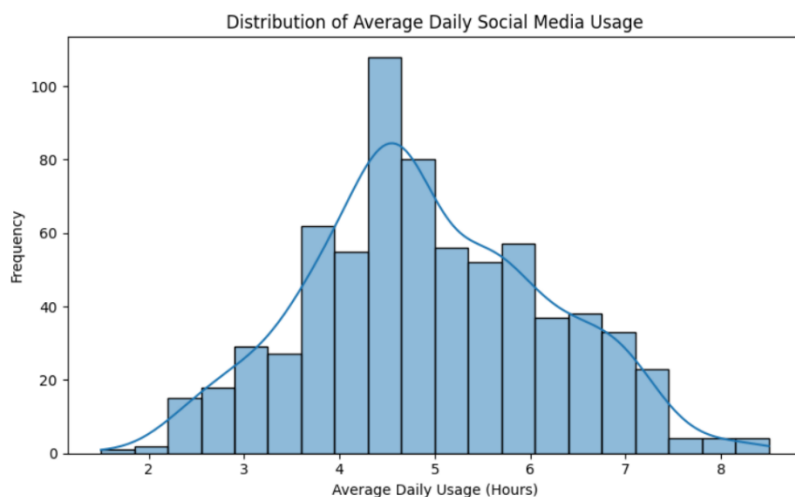
```
# Remove exact duplicate rows
df_clean = df.drop_duplicates()

# Optional: reset index after dropping duplicates
df_clean.reset_index(drop=True, inplace=True)
```

```
print(f"Number of duplicate rows: {df.duplicated().sum()}")
```

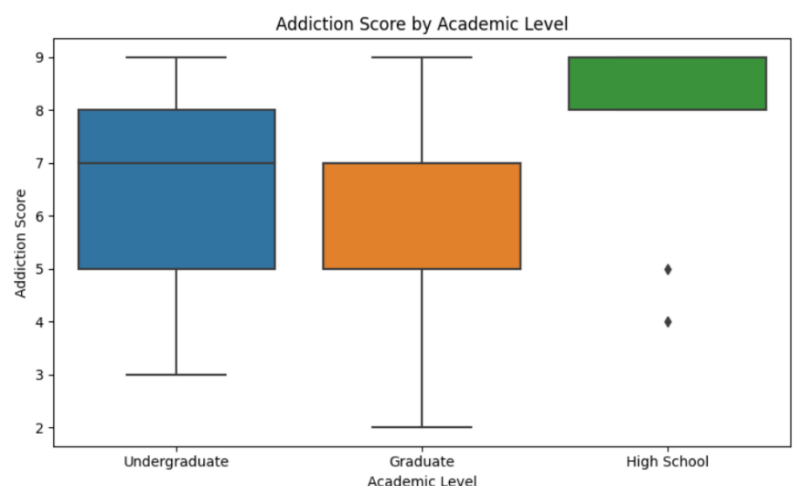
Number of duplicate rows: 0

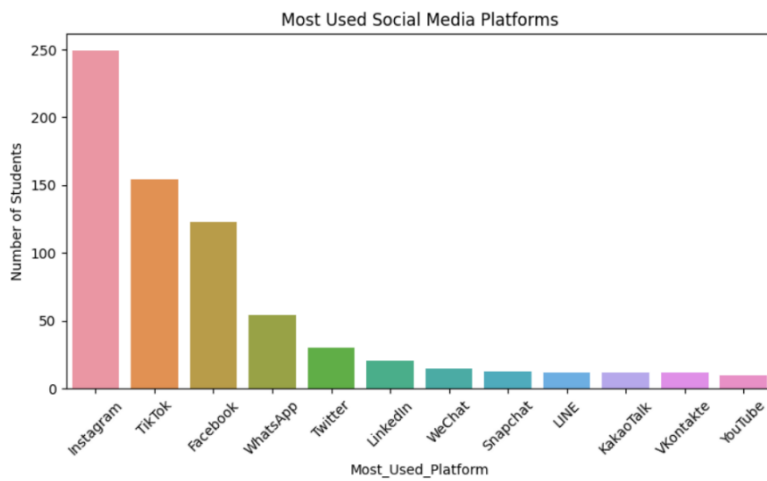
First, we remove any exact duplicate rows from the DataFrame and reset the index to ensure the row numbering is sequential and clean. Next, we verify that duplicates were successfully removed by counting and printing the number of duplicated rows, which confirms no repetitions remain.



This graph shows how many students use social media each day and for how many hours. The bars tell us how often different amounts of time happen, and the smooth line shows the overall trend of usage. It helps us see which daily usage times are most common.

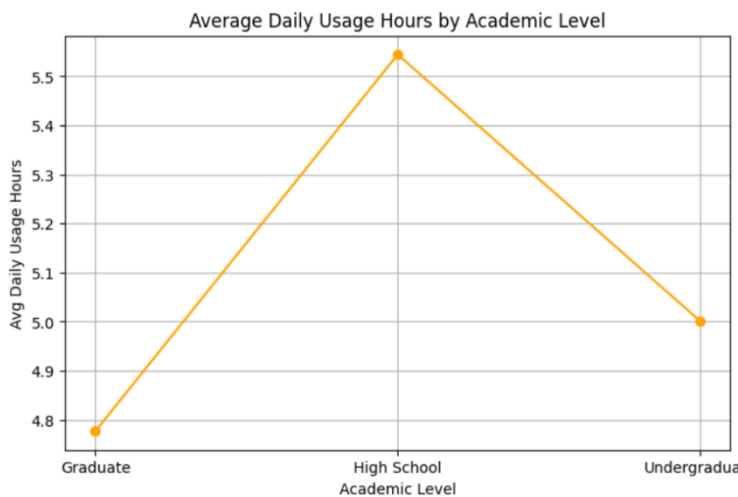
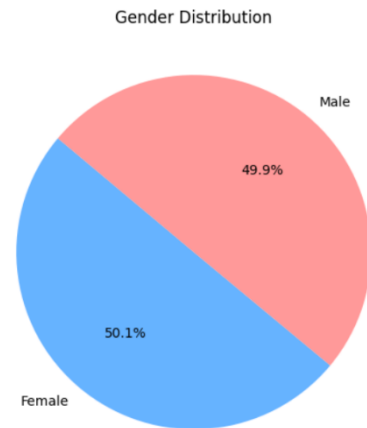
This box plot shows how the social media addiction scores change for students at different academic levels (High School, Undergraduate, Graduate). It helps us see if addiction is higher or lower in each group and how much the scores vary.





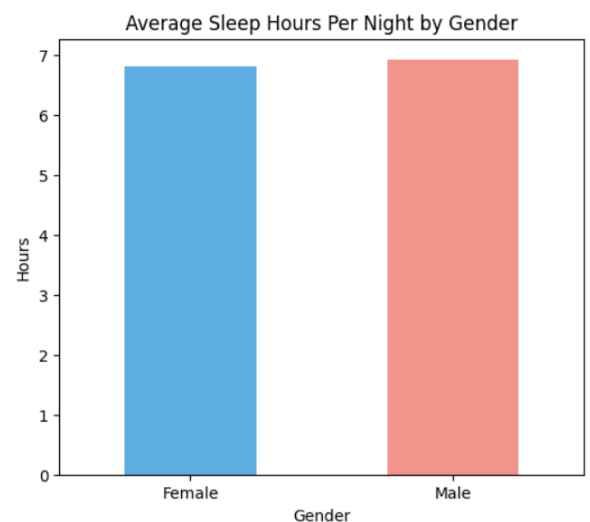
This bar chart shows which social media platforms are most popular among the students. The height of each bar tells how many students use each platform the most.

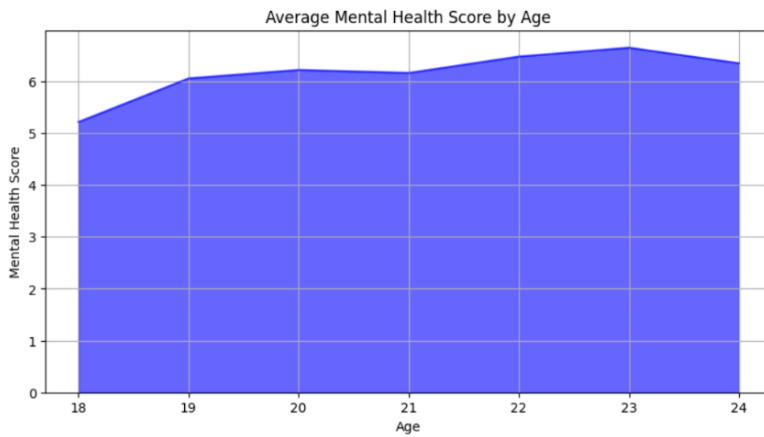
This pie chart shows the percentage of students by gender. Each slice represents how much of the group is male or female, with the percentages displayed on the chart.



This line chart shows the average daily hours students spend on social media, grouped by their academic level. It helps us see if usage changes between high school, undergraduate, and graduate students.

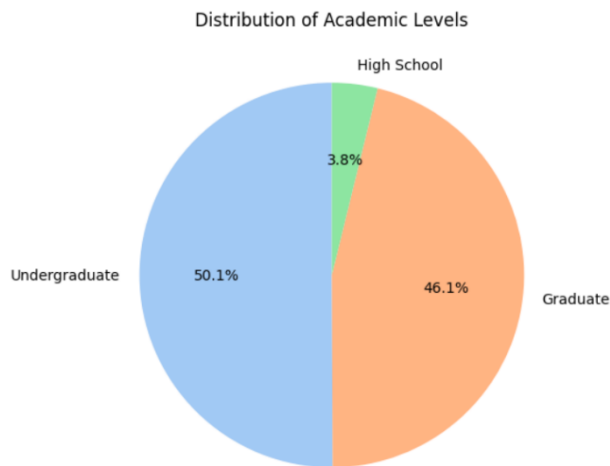
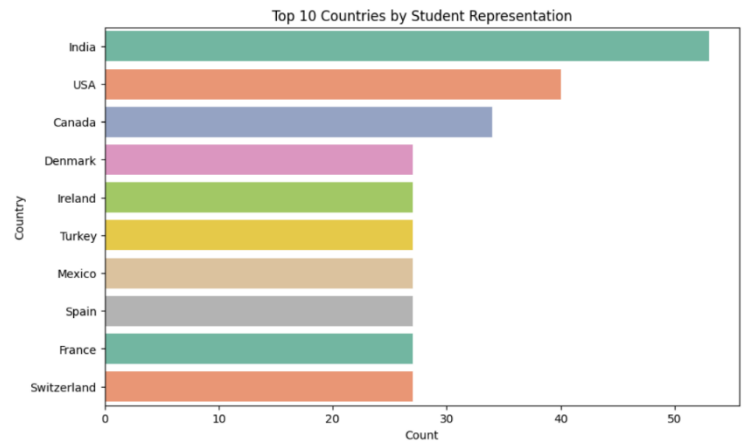
This bar chart shows the average number of hours students sleep each night, separated by gender. It helps us compare if males and females get different amounts of sleep.





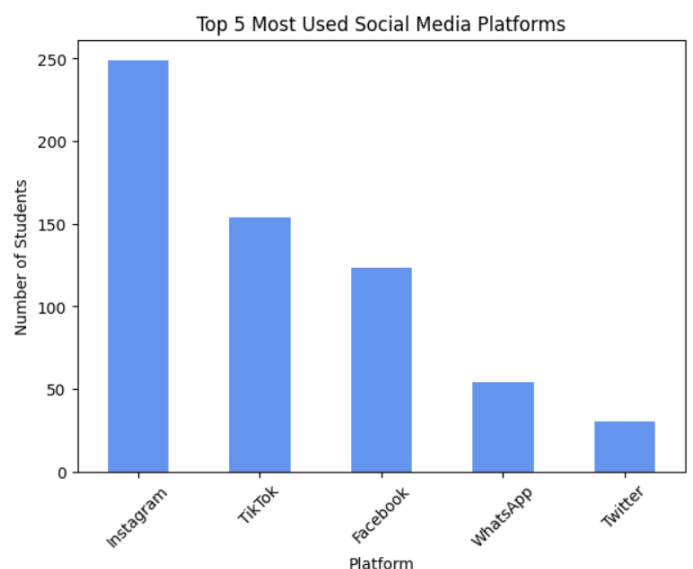
This area chart shows how the average mental health score changes with age. It helps us see if mental health tends to improve or drop as students get older.

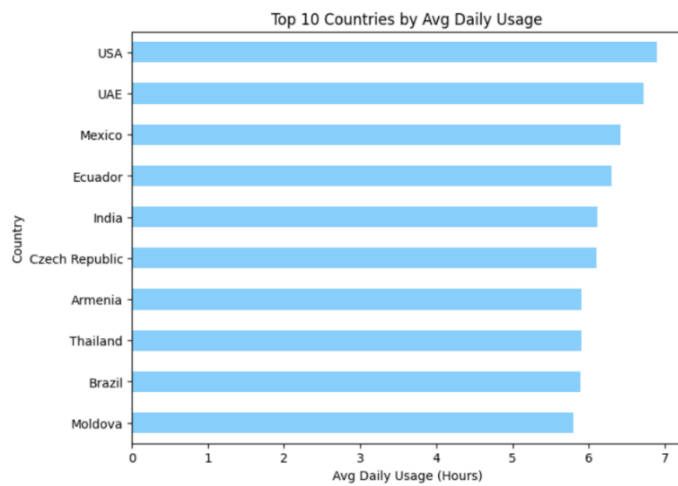
This bar chart shows the top 10 countries with the most students in the survey. It helps us understand which countries have the highest number of participants.



This pie chart shows how students are divided by their academic level — High School, Undergraduate, or Graduate. The percentages tell us how many students are in each group.

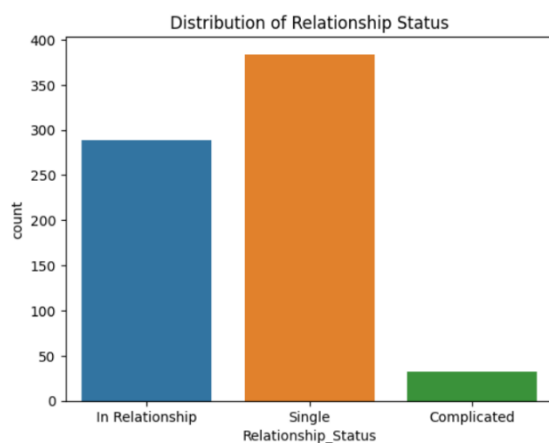
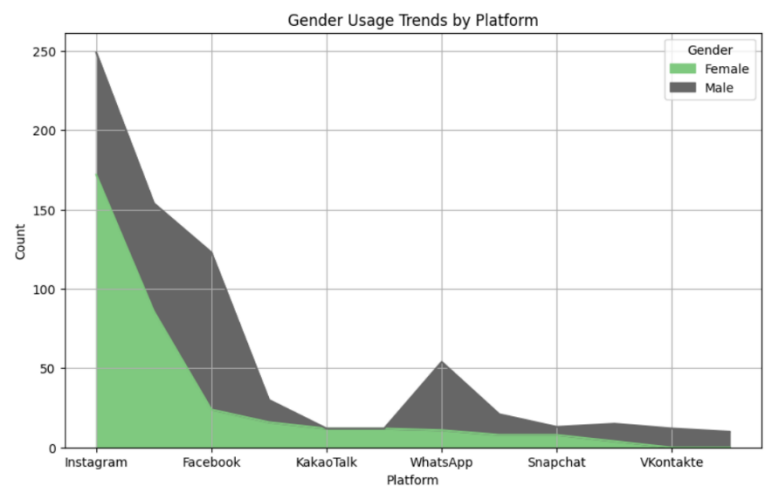
This bar chart shows the top 5 social media platforms that students use the most. The height of each bar tells how many students prefer each platform.





This chart shows the top 10 countries where students spend the most average hours per day on social media. The longer the bar, the higher the average daily usage in that country.

This chart shows how many male and female students use each social media platform. The colored areas represent the number of users by gender for each platform, helping us see which platforms are popular with each gender.



This bar chart shows how many students are in each relationship status category (Single, In Relationship, Complicated). It helps us understand the most common relationship situations among the students surveyed.

This stacked bar chart shows how students' relationship statuses (Single, In Relationship, Complicated) differ by their social media addiction levels (Low, Medium, High). Each bar represents one addiction level, and the colors inside the bar show the share of each relationship status within that group.

